# Lecture 18

Corey Powell

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# Midterm 2

- Midterm 2 will be in-class on Wednesday, November 11.
- The labs this week with cover an optional topic that will not be on homework assignments, quizzes, or exams.
- Midterm 2 covers probability:
  - Chapters 13 - 17 in the text,
  - Lectures 9 - 15 and associated review exercises,
  - Homework Assignments 5-7, and
  - R material from lectures and labs related to probability and simulation.

# R Code on Midterm 2

- Midterm 2 may ask for R code.
  - Calculate binomial or hypergeometric probabilities.
  - Simulate random variables.
  - Estimate probability, expected value, or standard error by simulation.
  - Use normal approximation to estimate probability.

# Lecture 18

1) A Comparison of Survey Methods
2) Brexit
3) Five Thirty Eight and the 2022 Midterm Elections
4) Reading Summary - Chapters 21 and 23
5) Making Confidence Intervals: Examples
6) Misinterpretation
7) Proper Interpretation
8) Why Do Confidence Intervals Work?
9) A Caveat

Read Chapters 6 and 24 for next Monday.

# A Comparison of Survey Methods

# Convenience Sample

- A generic term we will use for any non-random sample
  - Polling people you encounter on the street would be a classic example.
  - Inviting everyone in class to fill out an online survey would be another example.
  - The big concern is that a convenience sample is not going to be representative.

# Quota Sampling

- A specific type of non-random sample
- Requires that for certain variables (age, race, gender, etc) the sample is representative of the population.
- For other variables, the sample might not be representative at all.

# Simple Random Sample

- Essentially drawing without replacement from entire population
- Requires everyone to be known in advance
- Rarely practical, or even possible

# Cluster Sample

1) Divide the population into groups, often geographic.
2) Select some groups at random.
3) Sample the selected groups using:
   - Simple random sampling
   - Cluster sampling
     - Divide the selected groups into subgroups.
     - Select some subgroups at random.
   - Another sampling method

- A **multistage** sample has multiple rounds of cluster sampling.

# Cluster Sample vs Simple Random Sample

- A cluster sample is much more feasible than a simple random sample.
- A cluster sample has a higher standard error than a simple random sample.

# Stratified Sample

1) Divide (stratify) the population into groups, such as by age, race, gender, or other important variables.
2) Take a simple random sample of each group, proportional to the size of the group.

- The overall sample is therefore representative of the population in terms of the variables used in the stratification.
- Stratification requires that we know and have information on everyone in advance.
- Stratification lowers standard errors as compared to a simple random sample.

# Quota Sampling vs. Stratified Sampling

- Both require that the sample is representative of the population for certain key variables such as age, race, gender, etc. . .
- However,
  - stratified sampling incorporates randomization by requiring a simple random sample within each group, while
  - quota sampling does not incorporate randomization.

# Cluster Sampling vs. Stratified Sampling

- Both divide the population into subgroups.
- However,
  - cluster sampling only selects and samples from some subgroups, while
  - stratified sampling samples from all subgroups.

# Brexit

# Brexit Polls

- A June 23, 2016 poll of $4,772$ British citizens found that 52% supported remaining in the European Union, while 48% supported leaving the European Union.
  - See Poll Link
- The simple random sample formula for the standard error gives $\overline{0.52 * 0.48/4772} = 0.007$.
- Based on this standard error, it would seem highly unlikely that 50% or fewer British citizens support remaining.
  - $(0.50 - 0.52)/0.007 = -2.85$ standard errors

# Brexit Polls Part 2

- The standard error in the previous slide is deceptive for two reasons:
  1) The sample was not a simple random sample.
     - The standard error will be higher than the standard error for a simple random sample.
  2) The standard error ignores bias.

# Five Thirty Eight and the 2022 Midterm Elections

# Five Thirty Eight

- ABC eliminated Five Thirty Eight in March 2025.
- Five Thirty Eight used data and statistical analysis to try to answer questions in politics, sports, and public health.
- It was known for the accuracy of its' elections forecasts.
- It forecasted elections by summarizing poll results.
  - If one poll provides information, then multiple polls provide more information.
- Five Thirty Eight considered uncertainty in a much more sophisticated way than most political pundits.

# Five Thirty Eight and the 2022 Midterm Elections

- Five Thirty Eight created a model to forecast the results of the 2022 Midterm Elections.
- The model did a better job forecasting the election than most political pundits, who predicted a "red wave".
- The actual results were somewhat in the tail of what Five Thirty Eight predicted, however.
- The biggest difference between the forecast and what actually happened was in the Georgia Senate race.
- What happened?

# Five Thirty Eight and Poll Aggregation - Poll Quality

- Some potential problems with poll aggregation
  - Aggregating garbage polls will yield garbage information, and polls are not always transparent about their methods.
  - Five Thirty Eight tried to account for this by giving letter grade ratings to pollsters and weighting the polls based on the ratings.
  - These ratings were based on the pollster performance in recent elections.
    - They did not consider pollster methodology.

# Five Thirty Eight and Poll Aggregation - Poll Recency

- Five Thirty Eight weighted polls based on how close they were to the election.
- This was completely sensible.
- Partisan pollsters could use this to game the system by releasing polls right before the election.
  - Trafalgar Group is a Republican pollster who released lots of poll results right before the election.

# Five Thirty Eight and Poll Aggregation - Herding

- You expect to see variation in polling results.
- Polling results that are overly consistent should raise a red flag.
  - Pollsters sometimes use the term **herding** to refer to this phenomenon.
- There was perhaps some evidence of herding in Nevada.
- There was more herding in the 2016 presidential election polls.

# Review Questions

- See Review_questions17.pdf

# Reading Summary - Chapter 21

# Section 21.1: Introduction

- **Inference** is the process of generalizing from the sample to the population.
- With a simple random sample that is reasonably large, we can estimate the population percentage from the sample percentage.
- A key question is the likely size of the chance error. In other words, what is the standard error?
- It would be easy to calculate the standard error if we knew the contents of the box, but we don't!
- We can estimate the standard error by approximating the population percentage by the sample percentage.
  - This is called the *bootstrap method.*

# Section 21.2: Confidence Intervals

- A **confidence interval** is a range of values:

$$\text{sample percentage} \pm z \times \text{SE}$$

- The value $z$ determines the confidence level.
- If $z = 2$, the confidence level is approximately 95%.
- Make sure the normal approximation is valid before using this confidence interval.

# Section 21.3: Interpreting a Confidence Interval

- The confidence interval is random, **not** the parameter.
- The word "confidence" is a reminder of this fact.
- Different samples will result in different confidence intervals.
- However, the parameter is always the same.

# Section 21.4: Caveat Emptor

- The formulas in this chapter only apply to simple random samples.
- Most surveys don't have simple random samples.

# Section 21.5: The Gallup Poll

- Compares the actual Gallup Poll errors to those predicted by the simple random sample formulas.
- The real errors are much bigger than the formulas would predict!

# Reading Summary – Chapter 23

# Section 23.1: Introduction

- Focus of chapter: averages, as opposed to percentages
- Remember: averages and percentages are essentially the same thing.

# Section 23.2: The Sample Average

- The difference between the SD of the sample and the SE of the sample average
- We can use the SD of the sample to estimate the SD of the box.
  - This is the bootstrap method.
- Constructing confidence intervals for the population average

# Section 23.3: Which SE?

- A reminder of the standard error formulas for the sample sum, sample average, etc. . .

# Section 23.4: A Reminder

- The formulas only make sense in the context of a box model.
- It's easy to apply the formulas to many types of data, and people do.
  - The results mean nothing without a sensible box model.

# Making Confidence Intervals: Examples

# Procedure for Making Confidence Intervals

- Step 1: Set up a box model.
- Step 2: Calculate the sample average or percentage.
- Step 3: Calculate the approximate standard error
- Step 4: Look up in a $z$-table the value of $z$ corresponding to the confidence level.
  - It's roughly 2 for 95% (1.96 more precisely).
- Step 5: Calculate the confidence interval:
  $\text{sample average} \pm z \times \text{SE}$

# Example 1

- A box contains red marbles and blue marbles.
- The percentage of red marbles is unknown.
- You draw 200 marbles with replacement.
- 52 of the 200 marbles are red.
- What is a 95% confidence interval for the percentage of red marbles in the box?

# Example 2

- A high school has $1,200$ students.
- The principal wants to know the average GPA, and takes a simple random sample of 150 students.
- The average GPA in the sample is 3.12.
- The SD of the sample 1.15.
- What is an 80% confidence interval for the average GPA at the high school?

# Example 3

- A doctor wants to know what percentage of people in a certain city have a gene for cystic fibrosis.
- The doctor takes a simple random sample of 100 people.
- It turns out that 2 of them have a gene for cystic fibrosis.
- What is a 95% confidence interval for the percentage of people in the city with a gene for cystic fibrosis?
- How is this confidence interval problematic?

# Misinterpretation

# Common Misinterpretation

- It is easy to misinterpret confidence intervals.
- For example, after calculating a 95% confidence interval $(33.4, 34.8)$, someone might say that there is a 95% chance that the average of the box is in the range 33.4 to 34.8.
- This is wrong.
  - Basic reason: the average of the box is a non-random number that either *is* in the range $(33.4, 34.8)$, or *isn't*.
  - The chance statement doesn't make sense.

# Example

- Suppose that, unknown to the researcher, a box contains the tickets $\{1, 2, 3, ..., 1000\}$.
- The researcher takes a simple random sample of 100 tickets, and calculates a 95% confidence interval for the mean.
- We will simulate this in R.

# Confidence Interval (CI) function

```
#Takes a simple random sample SRS of n tickets
#from box and returns a 95% confidence interval
#for the mean of the box.
#Uses the correction factor for draws without
#replacement.
CI = function(box,n) {
  SRS = sample(box, n, replace=FALSE)
  sampleavg = mean(SRS)
  samplesd = sd(SRS)
  N = length(box) #The total population size
  se = sqrt((N-n)/(N-1)) * samplesd / sqrt(n)
  return(c(sampleavg - 2*se, sampleavg + 2*se))
}
```

# Confidence Interval Demo

```
set.seed(1)
box = 1:1000
mu = mean(box)
print(mu)
```

```
## [1] 500.5
```

```
print(sd(box))
```

```
## [1] 288.8194
```

```
print(CI(box,100))
```

```
## [1] 469.2802 576.2598
```

# What's Random, What's Fixed?

- The average of the box never changes. It's 500.5. It's not random.
- The confidence interval **is** random. It changes from sample to sample.
- It does not make sense to say that there is a 95% chance that 500.5 is in the confidence interval.

# Proper Interpretation

- A 95% confidence interval is a random interval.
- Suppose that we construct many simple random samples and calculate a 95% confidence interval for each of them.
- The 95% confidence means that roughly 95% of the confidence intervals will contain the mean of the box.
- In other words, 95% of the time you make a confidence interval, it will "work", giving you an interval that really does include the average of the box.
- The other 5% of the time it fails.
- You don't know, however, if a particular confidence interval will contain the mean.

# Why Do Confidence Intervals Work?

- Suppose that we have a box with an infinite number of tickets.
- The mean of the box is $\mu$ and standard deviation $\sigma$.
- Suppose that we take a simple random sample of size $n$ from the box.
- The Central Limit Theorem states that the sample mean $\bar{x}$ will be approximately normally distributed with mean $\mu$ and standard deviation $se = \frac{\sigma}{\sqrt{n}}$ if $n$ is large.
- This means that there is a roughly 95% chance that $\bar{x}$ is between $\mu - 2se$ and $\mu + 2se$.
- If we do some algebra with this inequality, we get a roughly 95% chance that $\mu$ is between $\bar{x} - 2se$ and $\bar{x} + 2se$.

# Why Do Confidence Intervals Work? Part 2

- Remember that $\bar{x}$ is a random variable in this formulation.
- The 95% chance means that we expect that $\mu$ is between $\bar{x} - 2se$ and $\bar{x} + 2se$ for roughly 95% of the simple random samples we create.
  - This is the **frequentist** interpretation of probability.
- This **does not** mean that there is a 95% chance that $\mu$ is between $\bar{x} - 2se$ and $\bar{x} + 2se$ for a particular simple random sample.

# Verification That Confidence Intervals Work Using Simulation

```
set.seed(1)
#Creates 1,000 simple random samples and CIs
#CIs is a table that holds the CIs
CIs = CI(box,100)
for (i in 1:999) {
  #Adds another CI to CIs
  CIs = rbind(CIs,CI(box,100))
}
#Percentage of CIs containing the mean of the box
print(100*mean(CIs[,1]<=mu & CIs[,2]>=mu))
```

```
## [1] 95.4
```

# A Caveat

- Consider the previous Example 3.
- Suppose that 2% of the population has cystic fibrosis.
- If we randomly sample 100 people from the population, then the number of people with cystic fibrosis will have a binomial distribution with $n = 100$ and $p = 0.02$.
- Let's simulate $1,000$ random samples of 100 people, calculate a 95% confidence interval for each sample, and determine the percentage of the confidence intervals containing 0.02.

# A Caveat Part 2

```
#Creates 1,000 simple random samples of 100 people
samples = rbinom(1000,100,0.02)
#The percentages of people with cystic fibrosis
pcts = samples/100
#Standard errors
ses = sqrt(pcts*(1-pcts)/100)
lower = pcts -2*ses
upper = pcts +2*ses
#Percentage of CIs containing 0.02
print(100*mean(lower<=0.02 & upper>=0.02))
```

```
## [1] 86.1
```

- What went wrong?

# Review Questions

- See Review_questions18.pdf on Canvas.